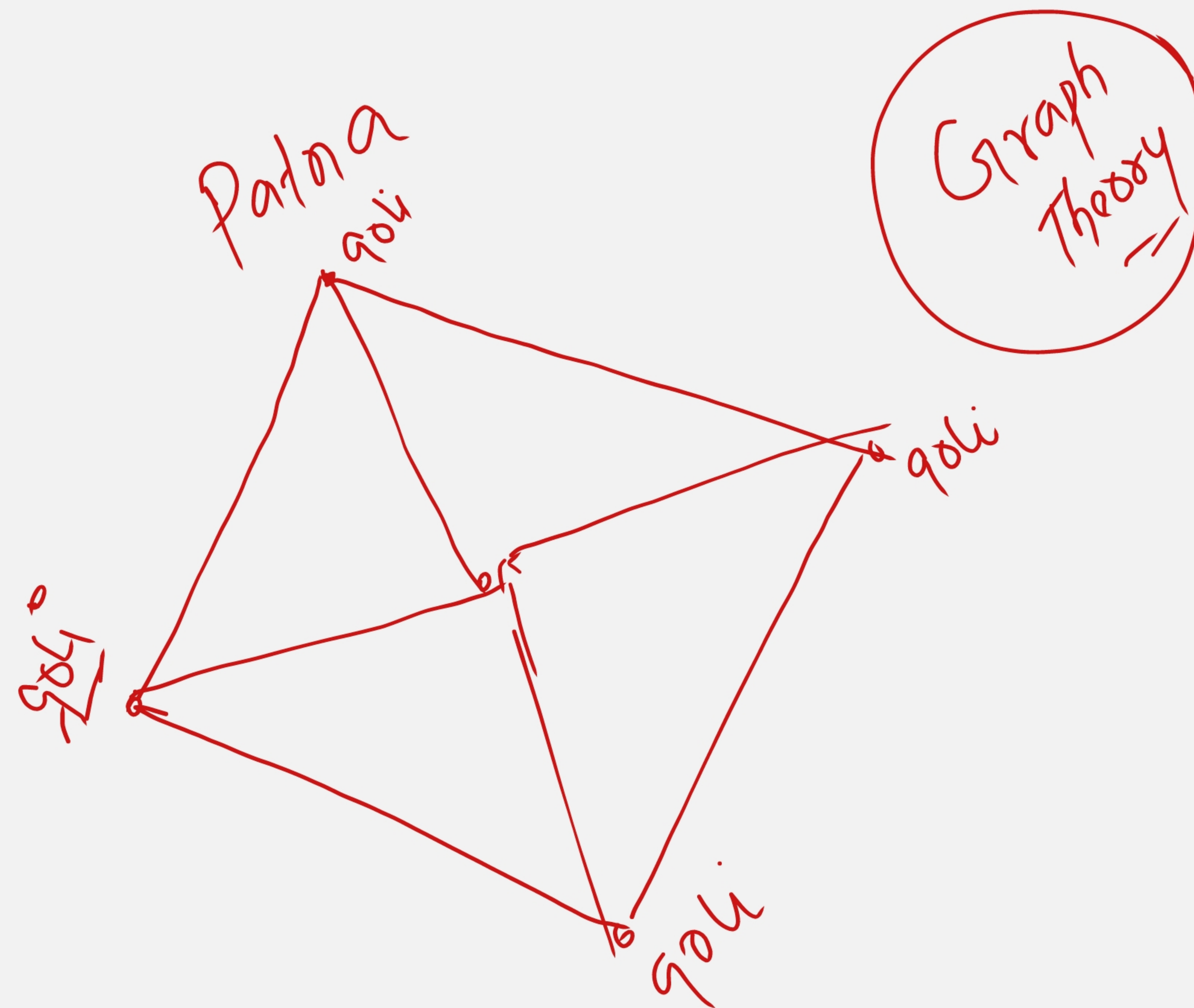


Topics to be Covered

Graph Theory

- 1) Definition
- 2) Types of Graph
- 3) Theorem of Simple Graph
- 4) Degree of vertex



Discrete mathematics

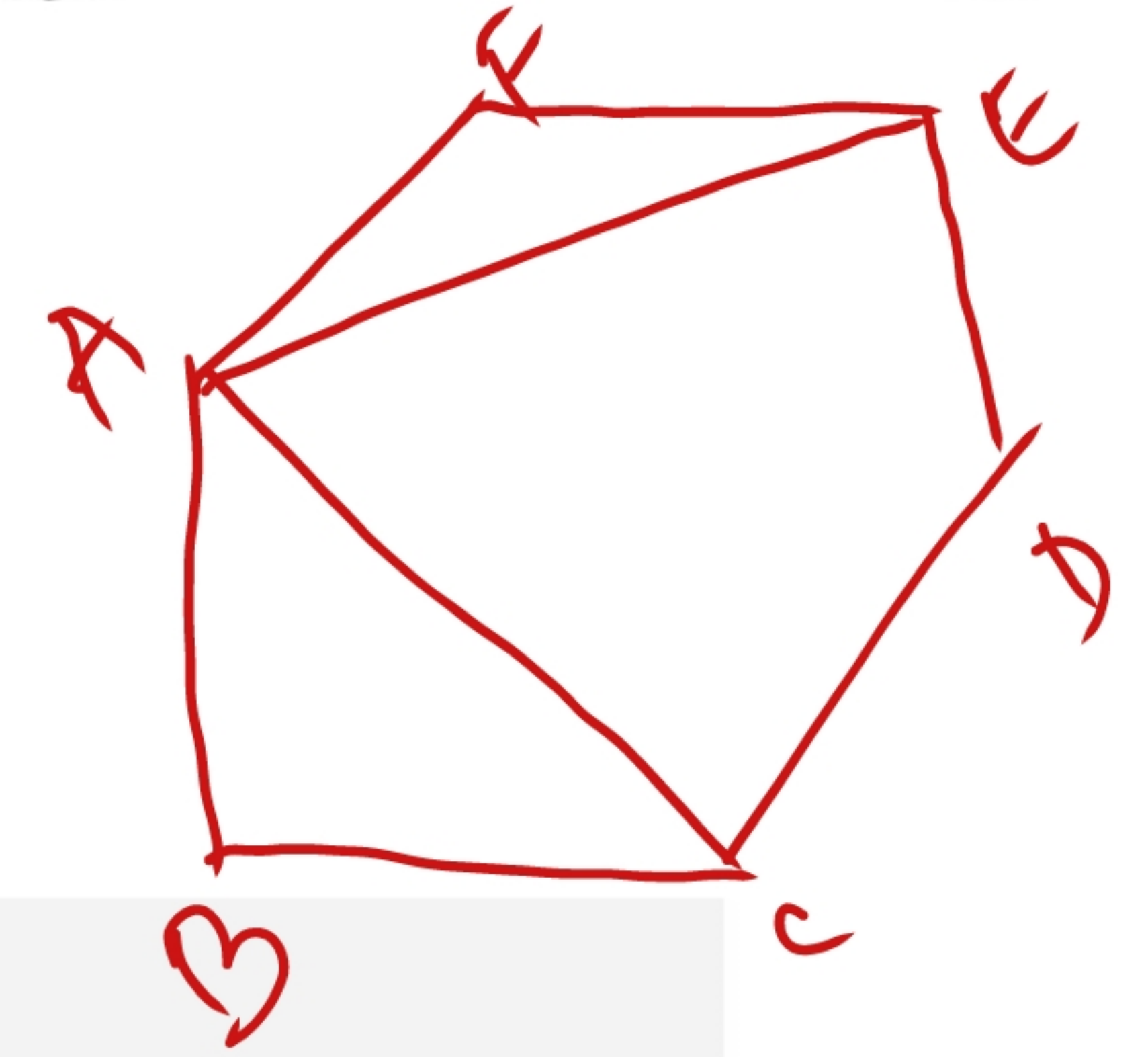
Graph Theory

Sanjay Kumar Sen
Assistant Professor at **Quad AI**

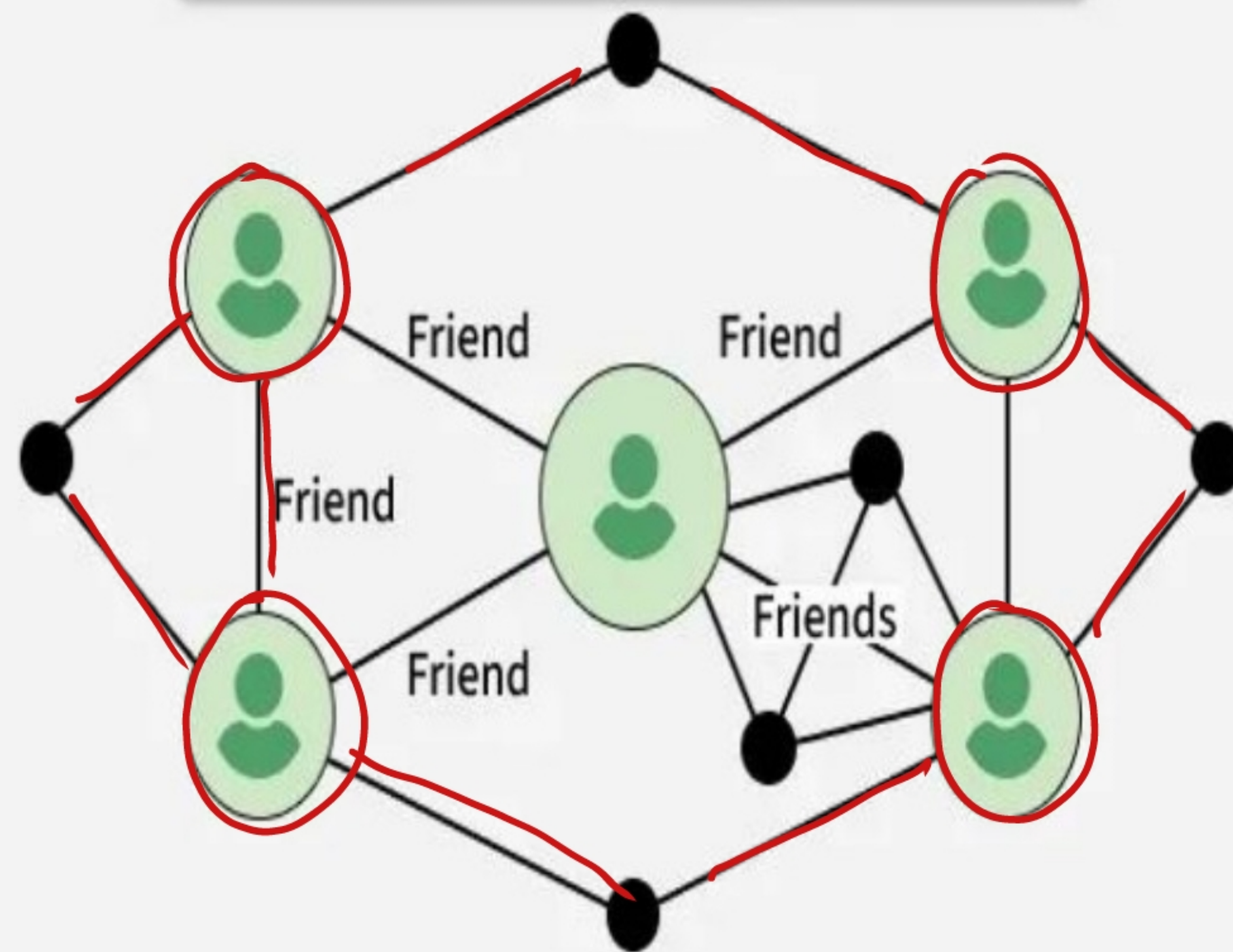


Introduction to Graph Theory

In many real-world situations, we often deal with a set of objects and the relationships between them. For example, cities connected by roads, people linked through friendships, or computers connected in a network. To study such relationships mathematically, we utilize graph theory.

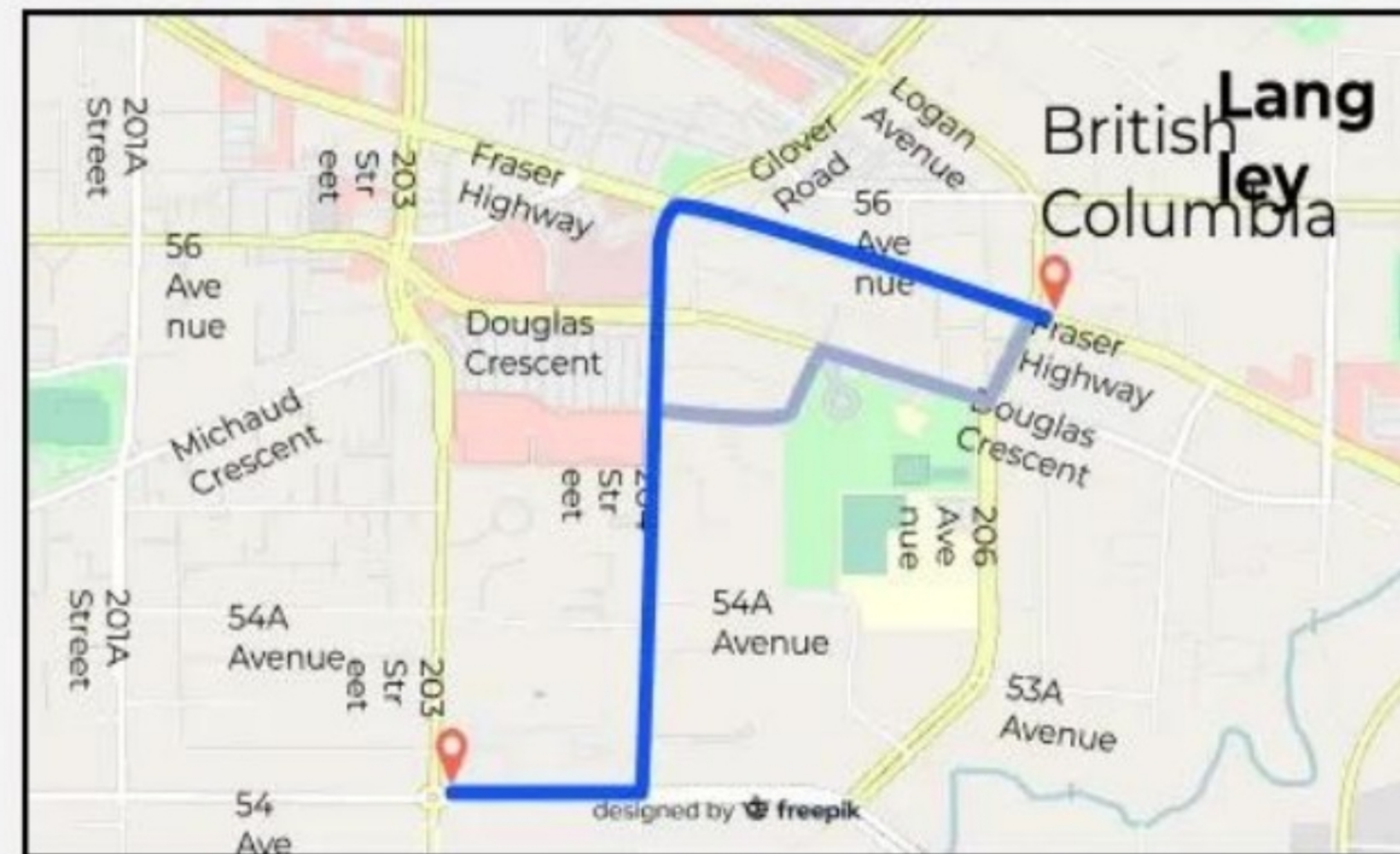


Social Networks



Graph theory behind social media connections ✓✓

Google Maps



Finding the shortest path using graph theory ✓✓



Basics of Graph

$\left\{ \begin{array}{l} \text{point / joint} \rightarrow \circ \rightarrow \text{vertex / vertices} \\ \text{line / branch} \rightarrow - \rightarrow \text{edge / edges} \end{array} \right\}$

Graph $G = (V, E)$

$\downarrow$ set of vertices

$\rightarrow$ set of edges

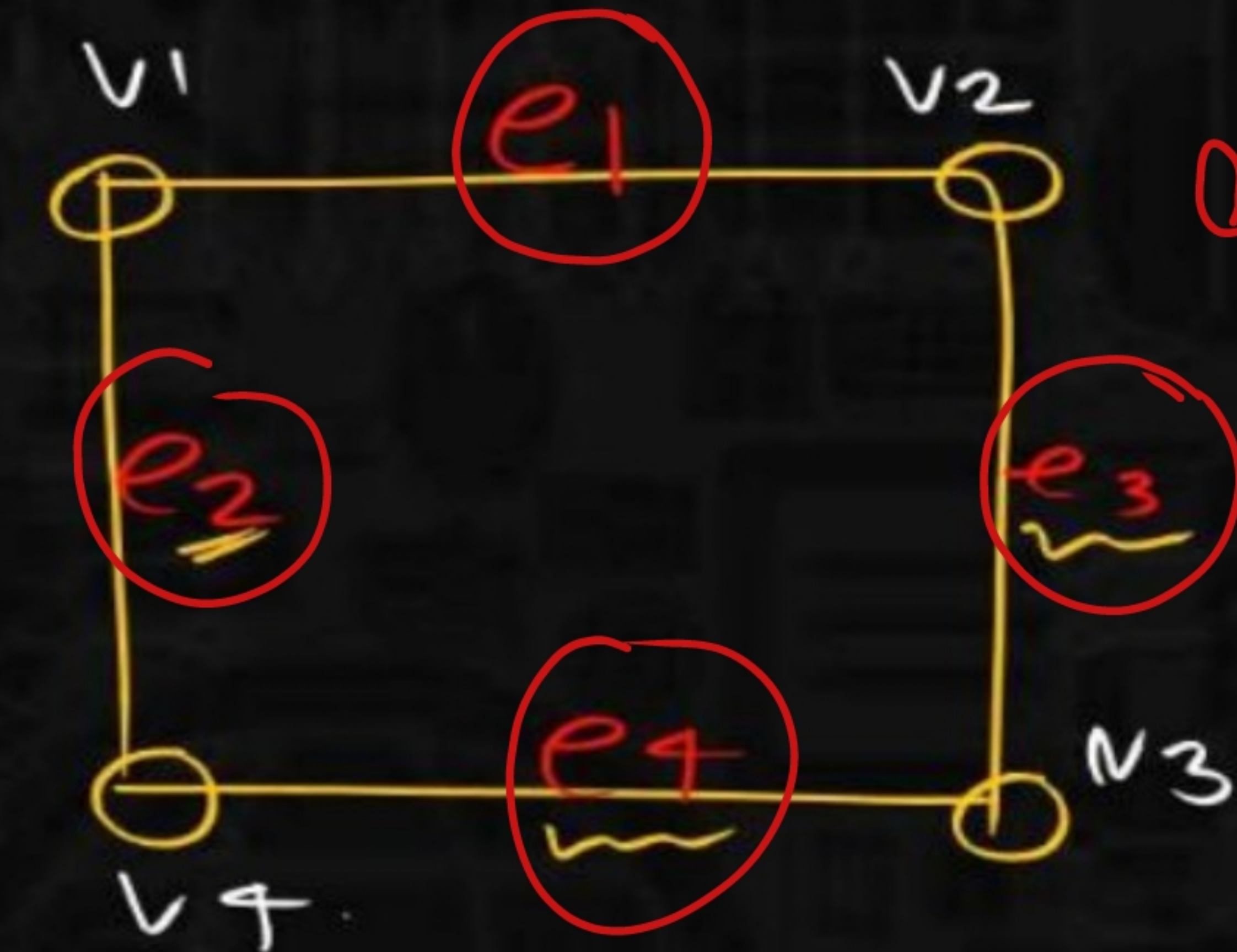


Basics of Graph

$$G = (V, E)$$

- ✓ 1. $V = \{v_1, v_2, v_3, v_4\}$
- ✓ 2. $E = \{e_1, e_2, e_3, e_4\}$

3. $\left\{ \begin{array}{l} \text{each } \underline{\text{edge}} \text{ is associated} \\ \text{with } \text{unordered pair of vertices.} \end{array} \right.$



$e_1 \rightarrow (v_1, v_2)$
 (v_2, v_1)

$$\begin{array}{l} e_1 \rightarrow (v_1, v_2) \mid (v_2, v_1) \\ e_4 \rightarrow (v_4, v_3) \mid (v_3, v_4) \end{array}$$

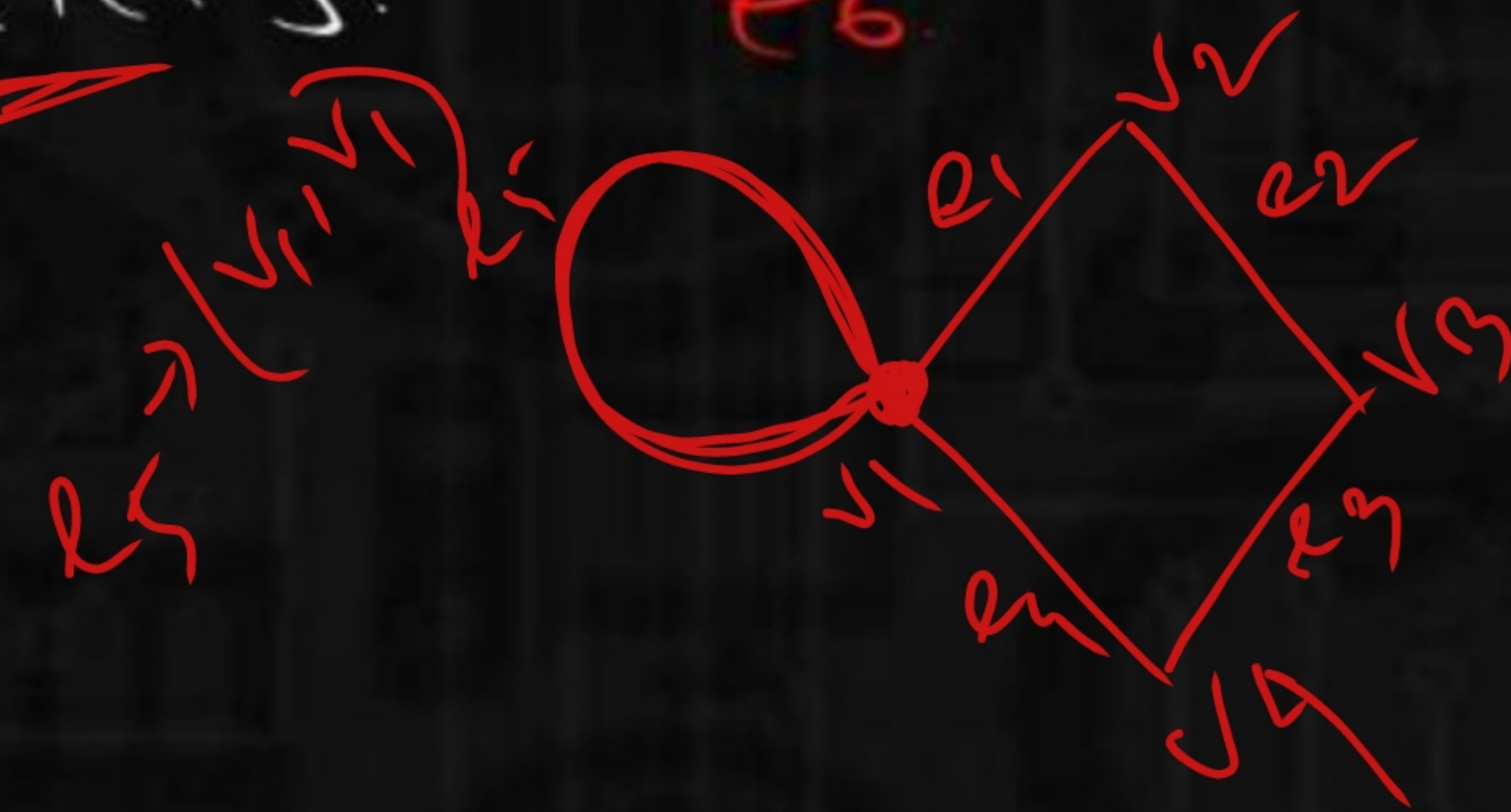
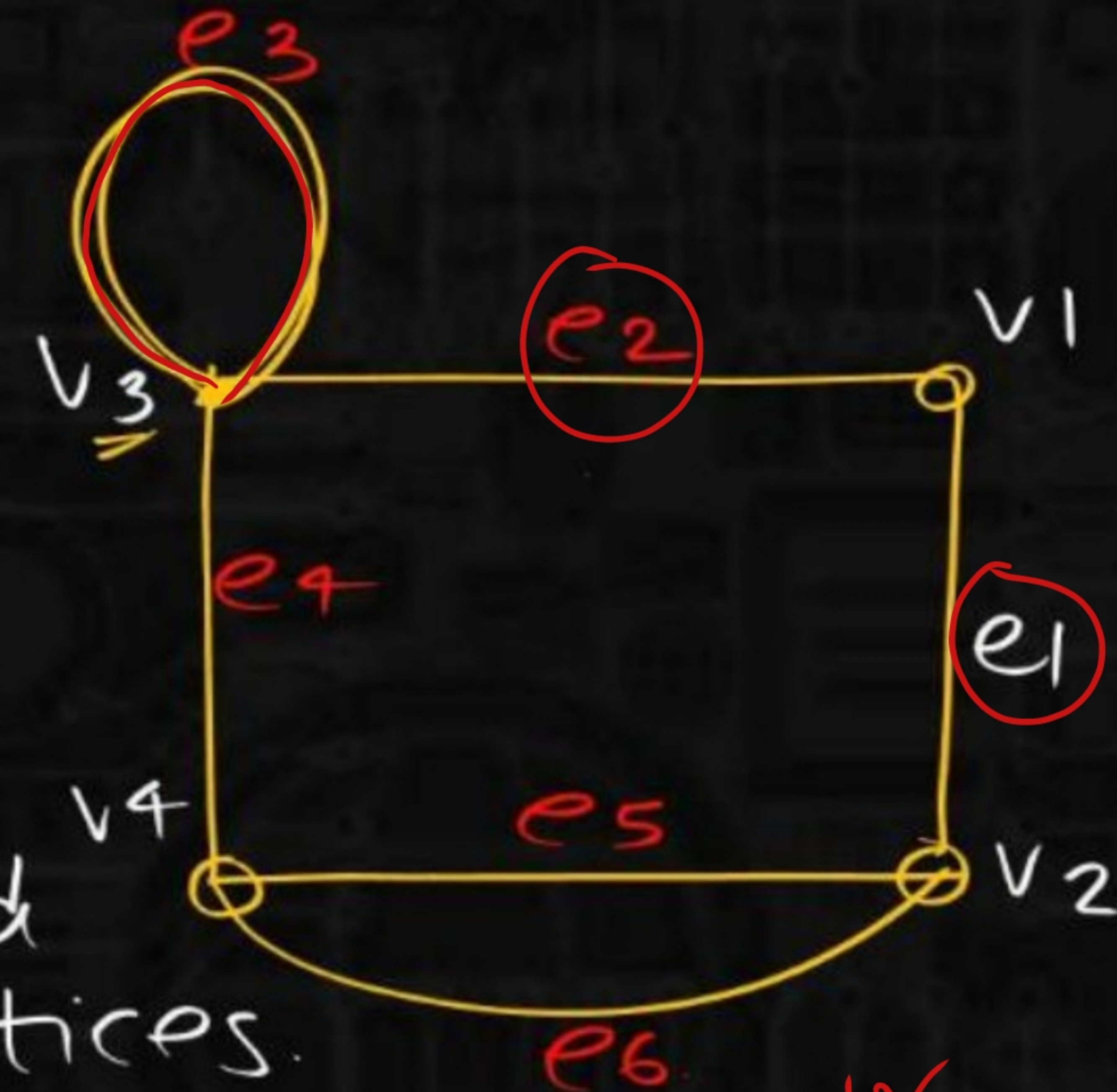


Basics of Graph

✓ end vertices: $e_1 \rightarrow (v_1, v_2)$

each edge is associated with unordered pair of vertices called as end vertices.

[loop/self loop]: $e_3 \rightarrow (v_3, v_3)$
if end vertices are same]



Basics of Graph

Degree/valency ($d(v_i)$)

i) no. of edges associated with vertex is degree of vertex

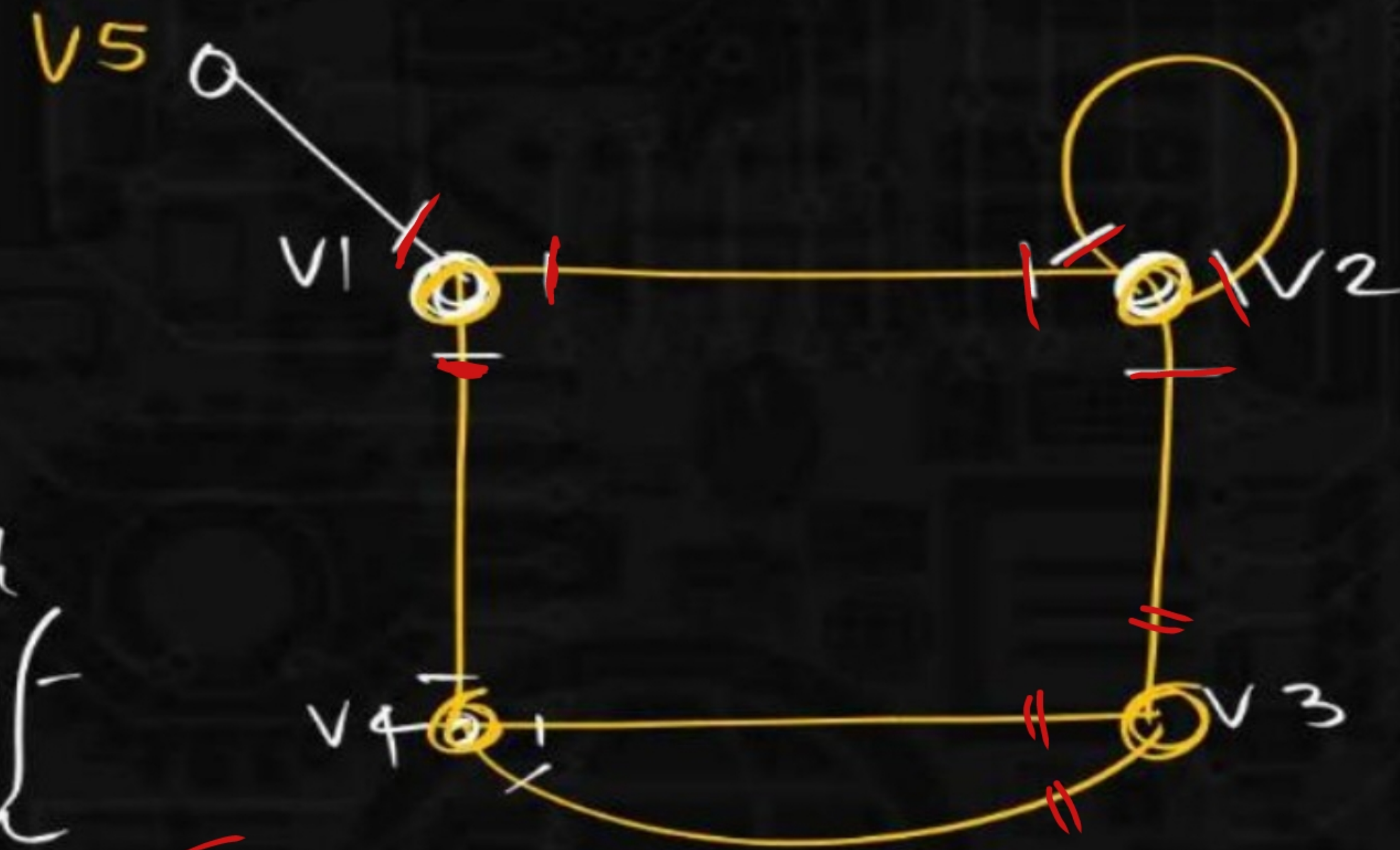
$$d(v_1) = 3$$

$$d(v_4) = 3$$

$$d(v_3) = 3$$

$$d(v_2) = 4$$

ii) Self loop is counted twice,



pendant vertex:
 $d(v_5) = 1$

{ Degree 1 is called pendant vertex

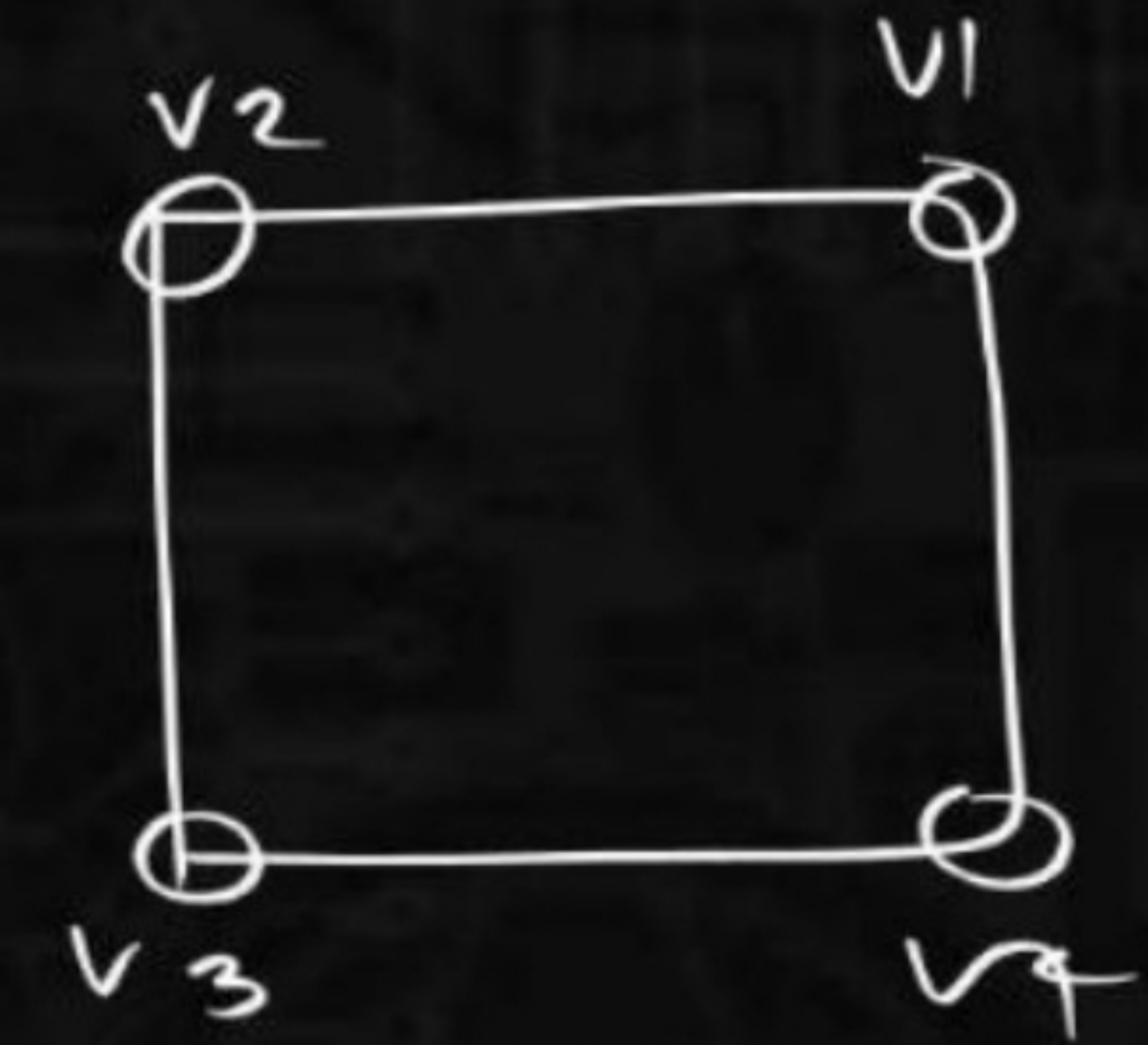


Basics of Graph

Isolated vertex:

$$D(v_s) = 0$$

if degree 0 is called isolated vertex.



$D(v_s) = 0$
 v_s

Null Graph : set of isolated vertices.

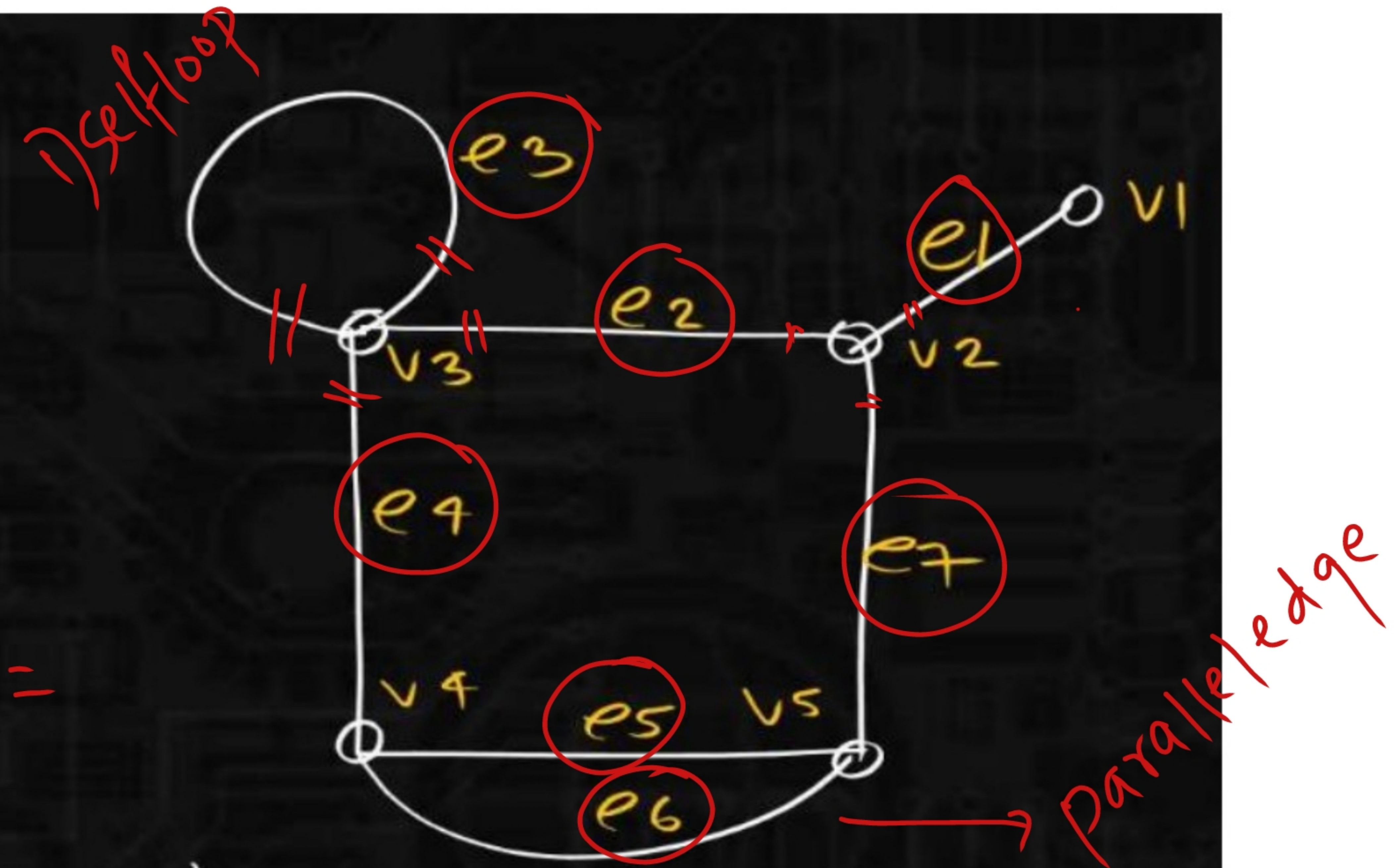


Basics of Graph

$$\begin{cases} d(v_1) = 1 \\ d(v_2) = 3 \\ d(v_3) = 4 \\ d(v_4) = 3 \\ d(v_5) = 3 \end{cases}$$

$$\sum d(v_i) =$$

$$\begin{aligned} & d(v_1) + d(v_2) + d(v_3) + d(v_4) + d(v_5) \\ &= 1 + 3 + 4 + 3 + 3 \\ &= 14 = 2 \times 7 \rightarrow \text{no. of edges} \end{aligned}$$



Conclusion

$$d(v_1) + d(v_2) + d(v_3) + d(v_4) + d(v_5) = 2 \times 7 \rightarrow \text{no. of edges.}$$

Thm: Sum of degrees of all vertices is equals to twice the no. of edge
 $\rightarrow \sum d(v_i) = 2e$ (even)

11 edges:

$e_1 \rightarrow (v_1, v_2)$

$e_2 \rightarrow (v_1, v_2)$

2 or more edges associated with same end vertices

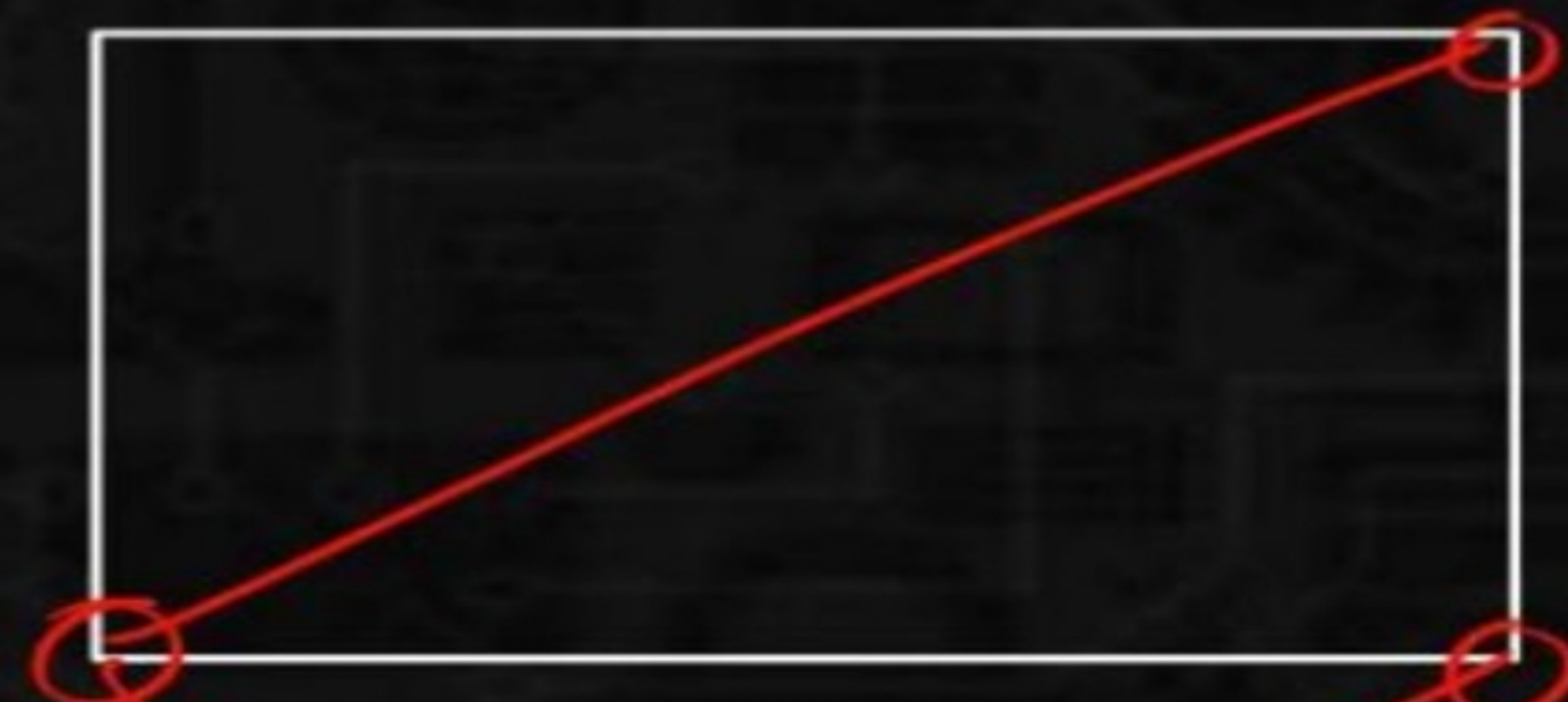


Basics of Graph

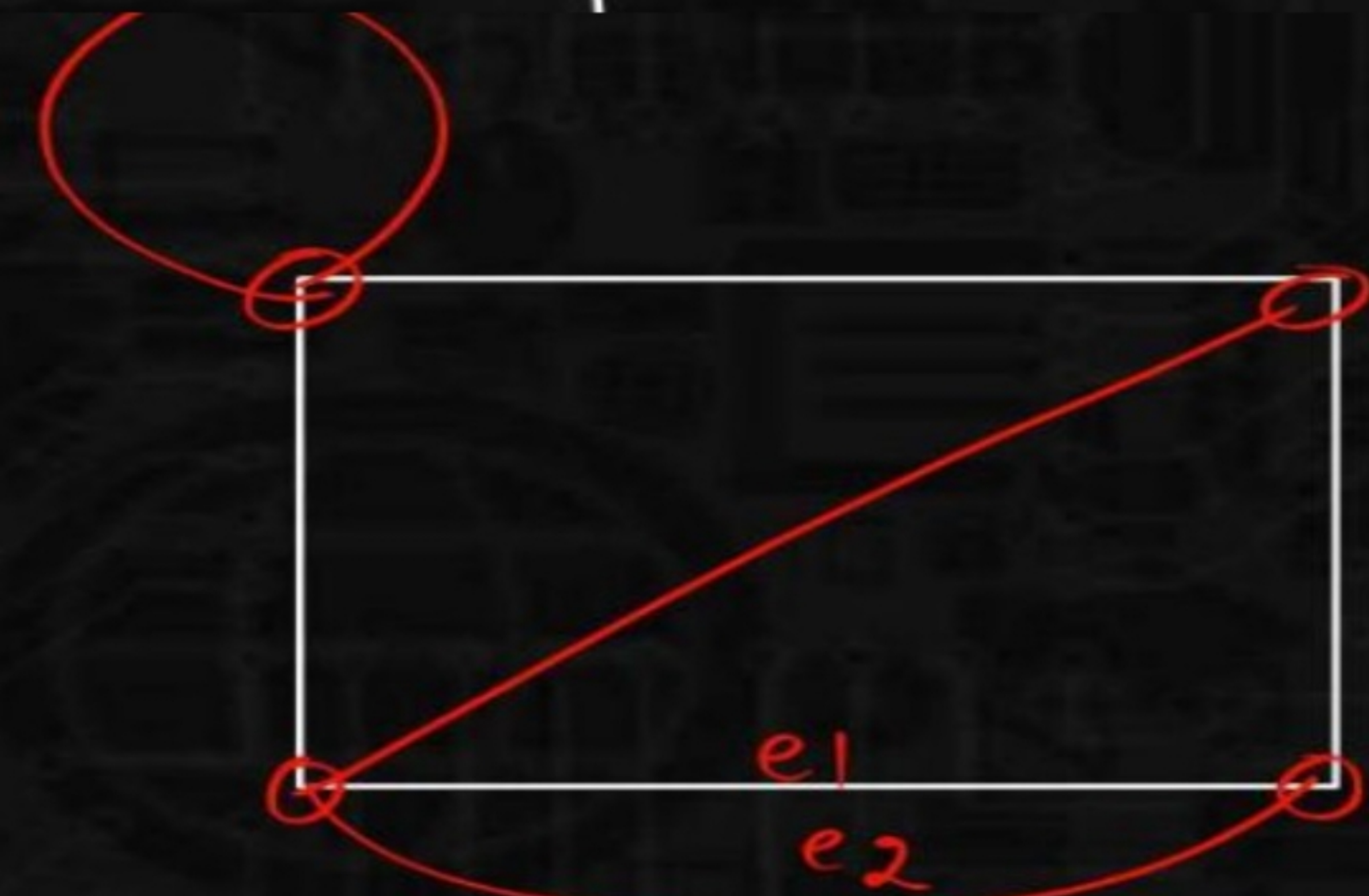
	loop	(Parallel)	edges
Simple Graph	X	X	X
multigraph	X	✓	✓
pseudograph	✓	✓	✓



Simple ✓

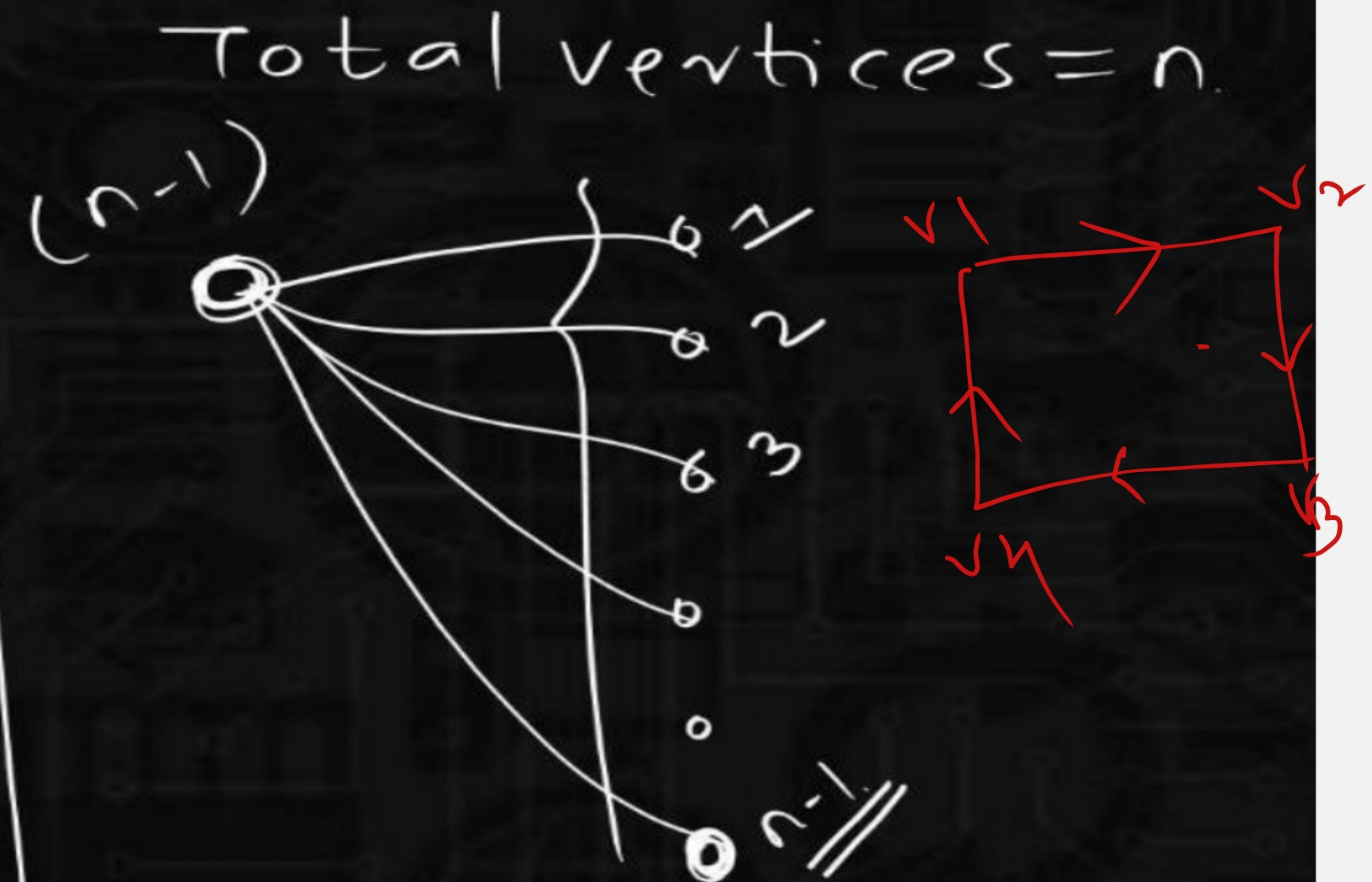
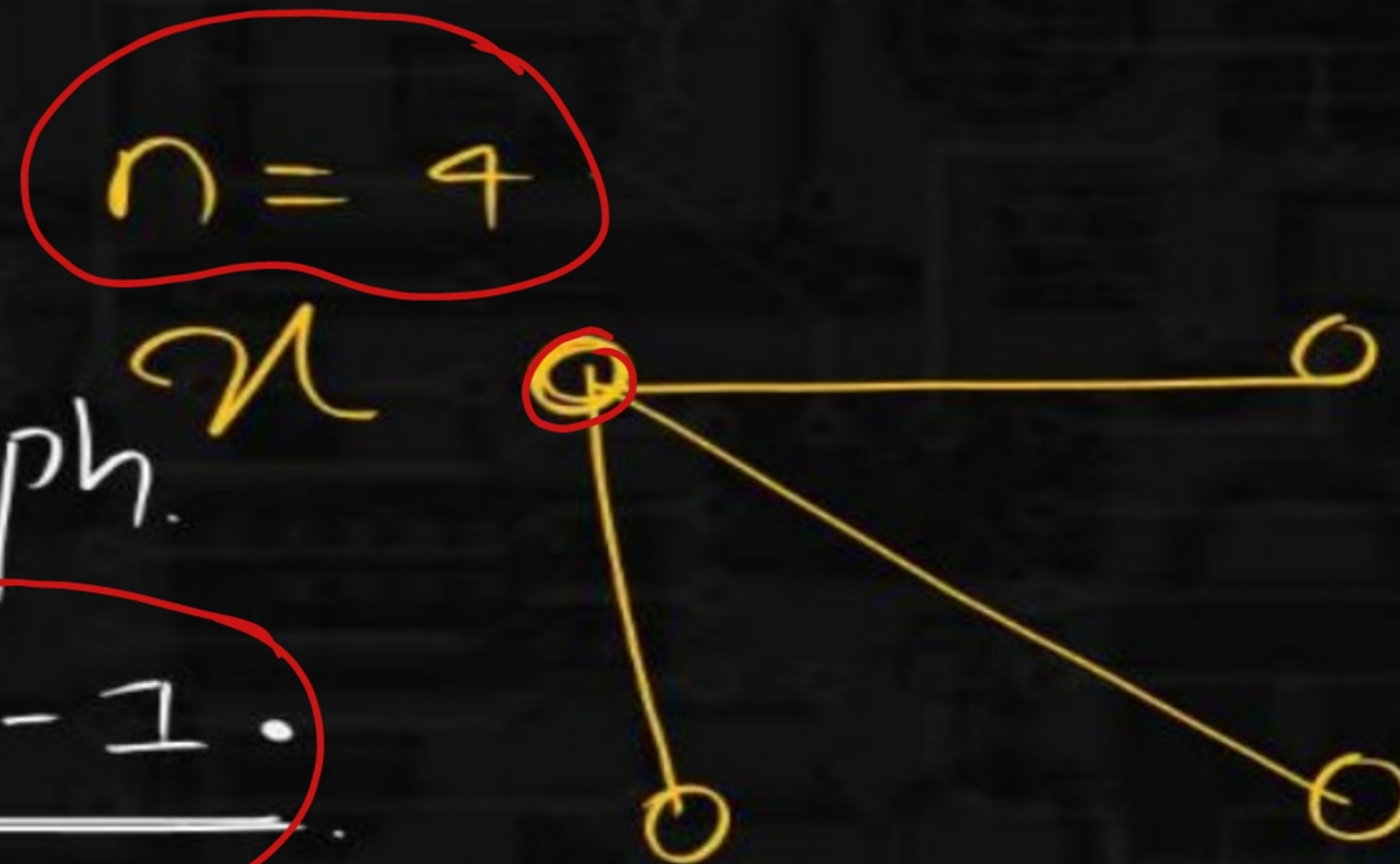


(multigraph)



Pseudograph.

Thm 3 ✓:
 maximum
 degree in
 simple graph.
 $\leq \underline{n-1}$.





Total vertices = $4 = n$.
Degree of
each vertex = $n - 1 (3)$

$$\sum d(v_i) = 2e$$

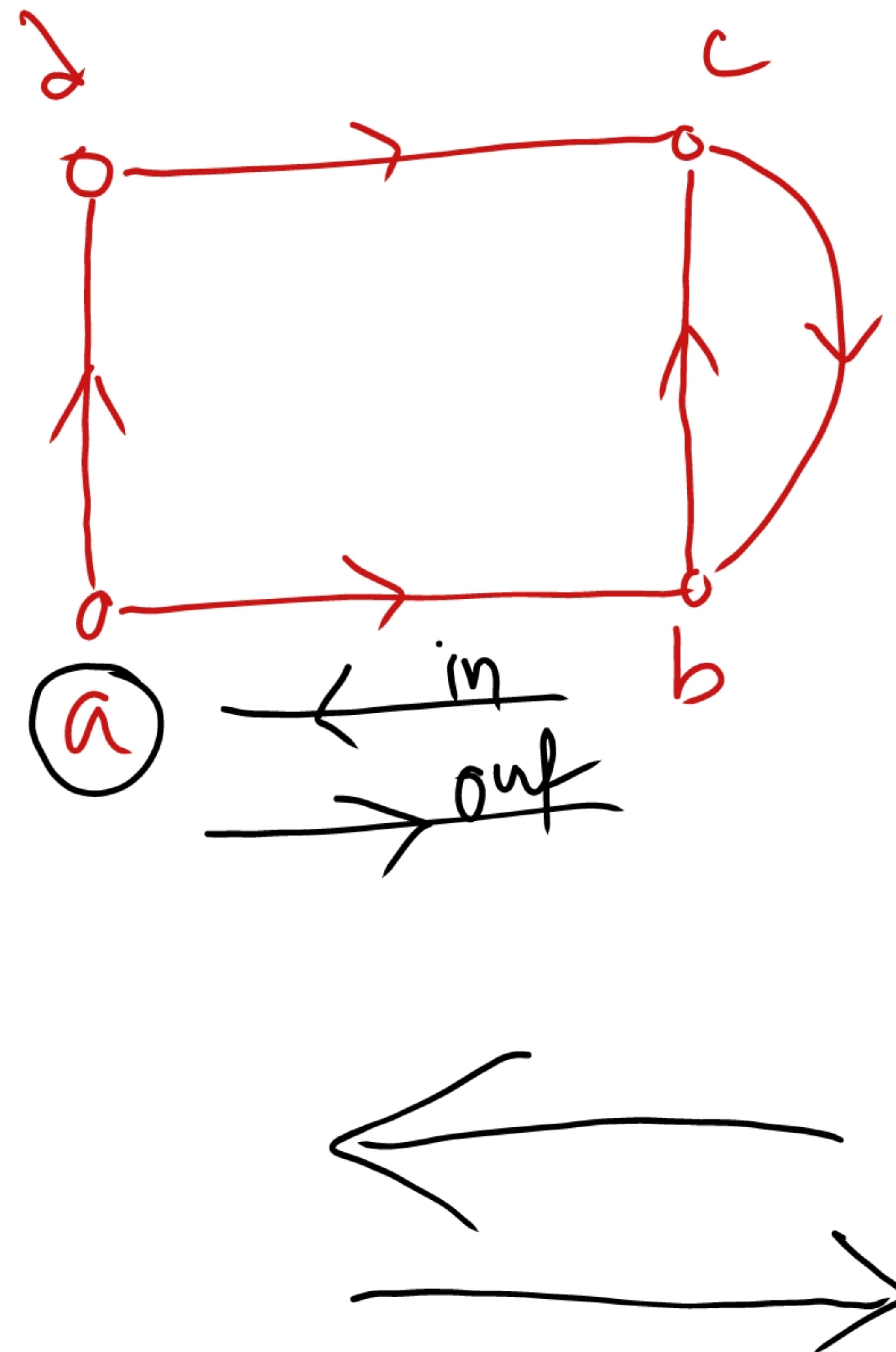
$$n - 1 + n - 1 + n - 1 \dots = 2e$$

$$n(n - 1) = 2e$$

$$e = \frac{n(n - 1)}{2}$$

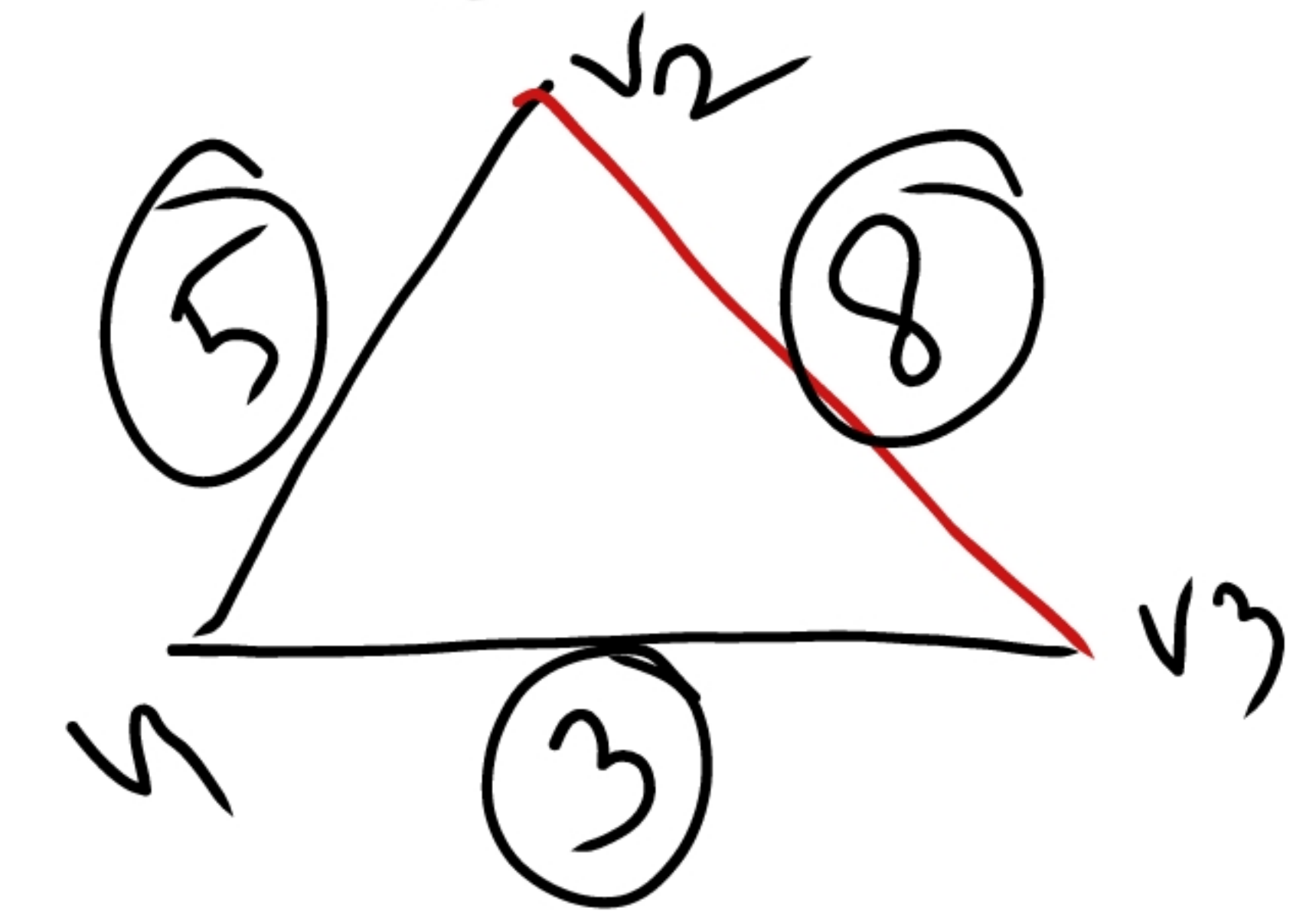
3
 $(n - 1) \times$
Total no. of edges = $\frac{n(n - 1)}{2}$

Indegree and out degree

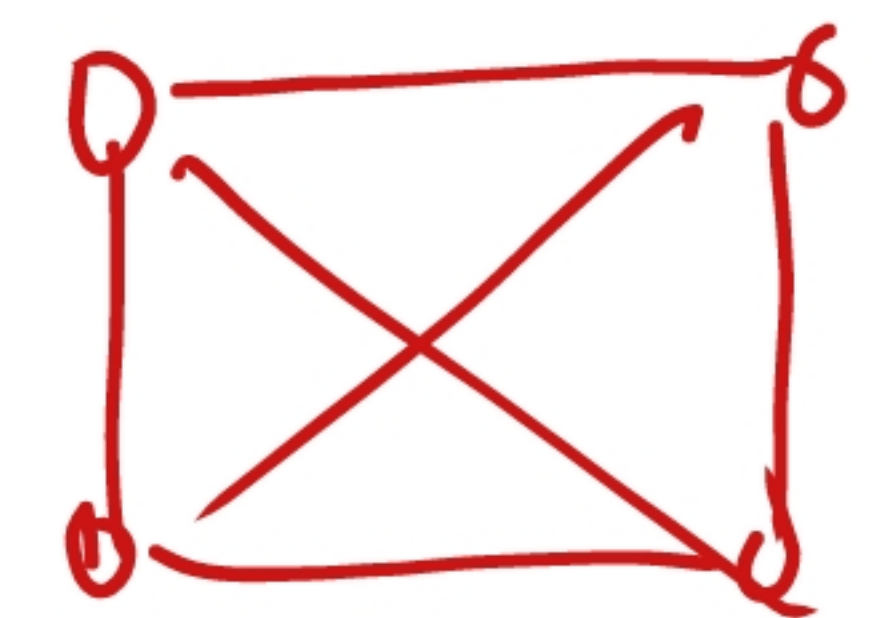


vertices	Indegree	Out-degree
a	0	2
b	2	1
c	2	1
d	1	1

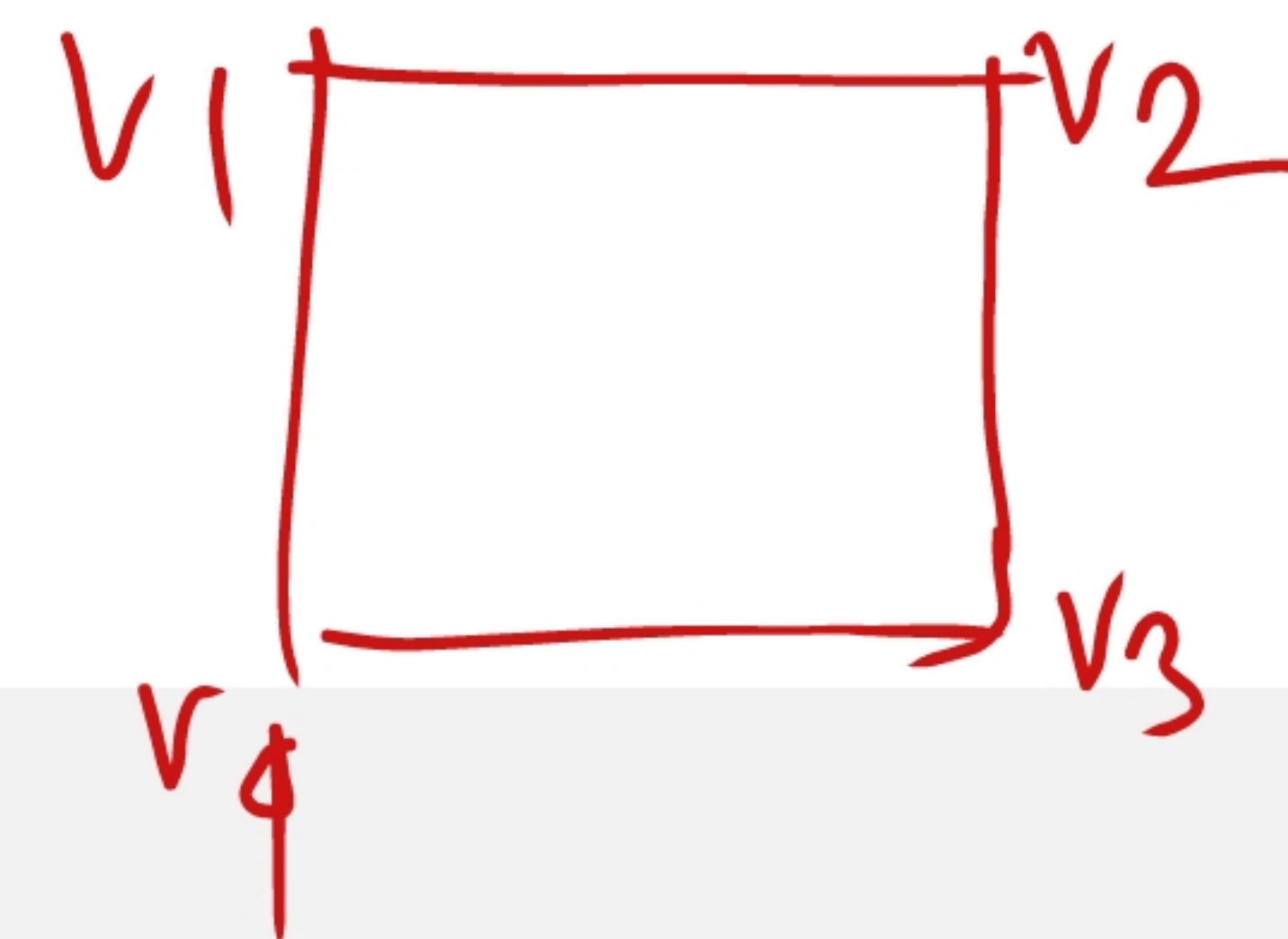
Weighted Graph :- A graph G in which some weight is associated to every edge of G .



Complete Graph :- A graph in which each vertex is connected to every other vertex.



Regular Graph :- Every vertex has same degree.

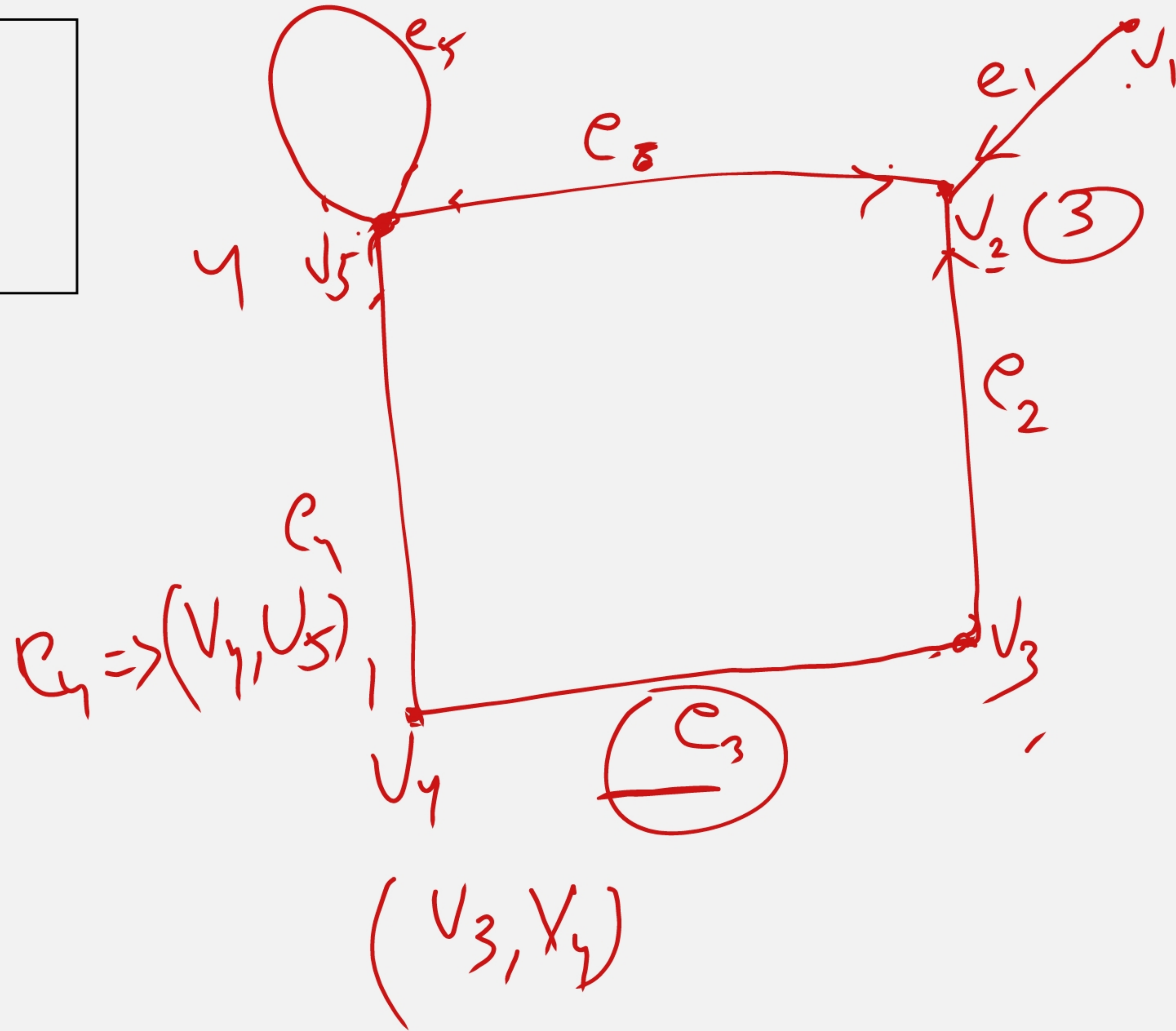


$$\left. \begin{array}{l} \deg(v_1) = 2 \\ \deg(v_2) = 2 \\ \deg(v_3) = 2 \\ \deg(v_4) = 2 \end{array} \right\}$$

Summary

$V \neq \emptyset$

V, E



Indegree Outdegree

Sum of Degree = $2E$



even

